581 Report — Auditing LLM Summaries: Sprint 3 Multi-Task Learning

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1 Abstract

Sprint 3 extends the Sprint 2 transfer learning pipeline by introducing a multi-task learning (MTL) objective. Instead of predicting only the `accuracy` label, models now jointly predict `accuracy` and `brevity` as a single combined label. We evaluate a traditional (Logistic Regression) and neural (MLP) MTL model, then compare several ensemble rules that combine Sprint 2 and Sprint 3 predictions. The Sprint 3 traditional MTL achieves the best pure Macro F1 of 0.421, outperforming the Sprint 2 weighted ensemble (0.372), primarily by recovering recall on the minority class 1. A confidence-gated hybrid of Sprint 2 and Sprint 3 predictions offers the best overall balance, reaching 0.750 accuracy and 0.415 Macro F1.

2 Introduction

Sprint 1 established TF-IDF baselines and Sprint 2 introduced GloVe transfer learning and soft-voting ensembles. A persistent challenge across both sprints was severe class imbalance, which caused models to over-predict class 2 with near-zero recall on classes 0 and 1. Sprint 3 addresses this by leveraging the `brevity` annotation as an auxiliary signal, with the hypothesis that brevity and accuracy are correlated and that jointly modeling both may improve minority class recovery.

Multi-task learning is a natural fit for this problem. Both labels are human annotations over the same summaries, so the model can share representations while simultaneously learning to distinguish multiple aspects of summary quality. Rather than modifying the input features as in Sprint 2, Sprint 3 changes the prediction objective itself, allowing the auxiliary task to act as a regularizer on the primary task.

3 Data

The same 400-entry CMV corpus and 320/40/40 train/dev/test split from Sprint 1 is used in this sprint, partitioned by thread and stratified by reviewer to prevent data leakage. No new data was collected. The key change is that both the `accuracy` (0–2) and `brevity` (0–2) annotation labels are now used during training, with `accuracy` remaining the primary evaluation target and `brevity` serving as the auxiliary task.

3.1 Auxiliary Task Selection

`brevity` was chosen as the auxiliary task over `sentiment` for two reasons. First, `brevity` has a healthier class spread than `sentiment`, which is heavily imbalanced, producing a more complete joint label space. Second, in a direct comparison, `accuracy` + `brevity` gave a better Macro F1 than `accuracy` + `sentiment` on the traditional multi-task setup. The intuition is that a summary capturing most key points concisely is often also more accurate, meaning brevity provides a complementary signal about overall summary quality that is closely related to the primary task.

3.2 Joint Label Construction

Labels from both tasks are combined into a single string of the form `acc_{accuracy}__brev_{brevity}` (e.g. `acc_2__brev_1`), producing a set of joint classes that the model predicts as a single multi-class problem. After prediction, the joint label is decoded back into separate `accuracy` and `brevity` predictions for evaluation. A `LabelEncoder` is fit on the training joint labels and applied consistently to the dev set.

4 Methodology

4.1 Feature Representation

Feature construction is identical to Sprint 2. The source text and summary are concatenated as `SOURCE: ...\nSUMMARY: ...` to form a unified input. TF-IDF features are extracted using unigrams and bigrams, with English stop words removed, a minimum document frequency of 2, and a vocabulary capped at the top 5,000 features. GloVe embeddings (`glove-wiki-gigaword-50`) are used to generate separate mean embeddings for the source and summary, which are then concatenated to form a 100-dimensional dense representation.

4.2 Traditional MTL Model

The traditional MTL model horizontally stacks the sparse TF-IDF features and the dense GloVe features, then trains a Logistic Regression classifier on the joint labels with `max_iter=2000`, `C=2.0`, and balanced class weights. After inference, the joint predictions are decoded and the `accuracy` component is extracted for evaluation against the gold labels.

4.3 Neural MTL Model

The neural MTL model reduces the TF-IDF features to 50 components via Truncated SVD, down from 100 in Sprint 2, and concatenates the result with the GloVe features before passing them to a single hidden layer MLP with 64 neurons, `max_iter=500`, and early stopping enabled. The same joint label training and decoding procedure is used as in the traditional model.

4.4 Ensemble Experiments

Six combination rules are evaluated using saved dev predictions from the Sprint 2 weighted ensemble and the Sprint 3 traditional MTL model. The Sprint 2 and Sprint 3 predictions are used as individual baselines. The min-label rule takes the lower of the two predictions. The brevity-gate rule uses the Sprint 3 prediction only when predicted brevity is low (≤ 1) and falls back to Sprint 2 otherwise. The confidence-gate rule uses Sprint 3 when its joint-label probability margin exceeds 0.329 and falls back to Sprint 2 otherwise. Finally, a three-model majority vote adds the Sprint 1 traditional baseline as a third voter, with ties broken by taking the lower label. The best rule is selected by Macro F1 with accuracy as a tiebreaker.

5 Results

5.1 Full Model Comparison Across Sprints

Model	**Acc.**	**F1**
S1 traditional baseline	0.650	0.336
S2 traditional transfer	0.625	0.366
S2 weighted ensemble	0.725	0.372
Neural MTL (MLP)	0.700	0.275
Traditional MTL (LogReg)	0.700	**0.421**
Confidence-gated (S2+S3)	**0.750**	0.415

Table 1: Dev performance across all sprints.

5.2 Ensemble Rule Breakdown

Rule	**Acc.**	**F1**
Sprint 2 only	0.725	0.372
Sprint 3 only	0.700	0.421
Min-label	0.700	0.421
Brevity-gate	0.700	0.384
Confidence-gate	0.725	0.372
Majority vote (3 models)	0.675	0.348

Table 2: Ensemble rules on the Sprint 3 dev set.

5.3 Per-class Breakdown

6 Analysis and Discussion

6.1 Multi-task Learning Improves Minority Class Recall

The traditional MTL model raises class 1 F1 from approximately 0.286 in the Sprint 2 ensemble to 0.455, driven by the regularizing effect of the brevity auxiliary signal. The joint label forces

Model	Cl.	P	R	F1
Trad. MTL	0	0.000	0.000	0.000
	1	0.455	0.455	0.455
	2	0.793	0.821	0.807
Neural MTL	0	0.000	0.000	0.000
	1	0.000	0.000	0.000
	2	0.700	1.000	0.824

Table 3: Per-class precision (P), recall (R), and F1 for both MTL models.

the model to distinguish more fine-grained combinations of accuracy and brevity, which indirectly discourages over-prediction of the majority class. The overall effect resembles regularization rather than a fundamental behavioural change: the model becomes less overconfident about class 2, which helps on borderline class 1 examples while still correctly retaining most class 2 predictions.

6.2 Neural MTL Collapses to Class 2

The MLP model predicts class 2 for every example, producing a degenerate Macro F1 of 0.275. This is consistent with the neural model's behaviour in Sprint 1 and Sprint 2. The most likely causes are the smaller feature space from reducing SVD components to 50, higher model capacity relative to dataset size, and the imbalanced joint label distribution overwhelming the minority signal. The multi-task objective does not help the neural model break out of the majority-class pattern the way it does for the linear model.

6.3 Ensemble Results and the Confidence-Gated Hybrid

No simple ensemble rule outperforms the Sprint 3 traditional MTL on Macro F1 alone. However, the confidence-gate rule produces the best overall balance, reaching 0.750 accuracy and 0.415 Macro F1 by using the Sprint 3 prediction when the model's joint-label probability margin is high and falling back to the Sprint 2 ensemble otherwise. This is the highest accuracy result across all three sprints and still outperforms the Sprint 2 weighted ensemble on Macro F1, making it the most well-rounded combined model. The three-model majority vote performs worst overall, suggesting the Sprint 1 baseline adds noise rather than a useful third signal.

6.4 Qualitative Error Analysis

Comparing the Sprint 1 traditional baseline against the Sprint 3 traditional MTL on dev predictions reveals a clear pattern in both the fixed and broken cases [?]. Among the fixed examples, `152_u` and `159_u` were cases where the baseline over-predicted class 2 and the MTL model correctly moved predictions down to class 1. Both summaries are fairly complete but not clearly strong enough for the top accuracy label, suggesting the brevity signal is acting as a regularizer. A third fixed example, `45_s`, shows the opposite direction: the baseline under-predicted at class 1 while the MTL model correctly moved it up to class 2, demonstrating that the joint label can also support upward corrections on summaries that are both accurate and complete.

Among the broken examples, `159_s` and `184_u` were cases where the baseline correctly predicted class 2 but the MTL model dropped predictions to class 1. The model's increased caution, while beneficial overall, leads to underprediction on summaries that are genuinely strong. The brevity signal may have pulled the joint prediction toward a lower accuracy class in these cases.

6.5 Class 0 Remains Undetected

No model across any sprint has successfully predicted class 0. This reflects both the extreme rarity of class 0 examples in the training set and the inherent difficulty of distinguishing "not accurate" summaries from "moderately accurate" ones using surface-level lexical and embedding features alone.

6.6 Data Limitations and Noise

The dataset is small and the `accuracy` label is imbalanced, which places a hard ceiling on how much any model can realistically improve. The annotations are subjective, reflecting human judgments of summary quality rather than objective ground truth. Furthermore, the prediction target is itself a rating of LLM-generated summaries, adding an additional layer of noise on top of already messy natural language data. Some instability across models and limited absolute gains are therefore expected consequences of the low-resource noisy setup rather than indicators of poor modeling.

7 Conclusion

Sprint 3 demonstrates that multi-task learning with `brevity` as an auxiliary task meaningfully improves accuracy classification, particularly for the difficult class 1. The traditional MTL model, with a Macro F1 of 0.421, is the best single model across all three sprints. The confidence-gated hybrid, at 0.750 accuracy and 0.415 Macro F1, is the best balanced combined result and the strongest overall model when both metrics are considered together.

Ensemble combinations of Sprint 2 and Sprint 3 models offer limited gains over the MTL model alone, suggesting that the MTL objective is itself the more effective integration strategy compared to post-hoc rule-based combination. Class 0 prediction remains an open problem that will likely require more data, oversampling, or a fundamentally different modeling approach. Future work could explore sentence-level embeddings such as SBERT, explicit class-rebalancing via SMOTE, or fine-tuning a small pretrained encoder directly on the annotation task.

References

[1] ChatGPT. Response to "Help brainstorm, debug, and revise project code and documentation for this project." OpenAI, 18 Apr. 2026. `https://chat.openai.com/`.

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[3] Claude Sonnet 4.6. Response to "Organise the following bullet points byfollowing the section headings and academic layout of the provided scientific report." Anthropic, 18 Apr. 2026. `https://claude.ai/code`.

[4] Claude Sonnet 4.6. Response to "This is a draft of a scientific paper. identify where the writing style is inappropriate for a scientific paper, and suggest corrections." Anthropic, 18 Apr. 2026. `https://claude.ai/code`.